

SURGE

Build Guide

All of the SMD components on the PCB have already been assembled. So, this is a real quick and easy build.

1. Start by placing the 2x5 power header. The placement of the power header is on the downside of the main PCB (the PCB with the square hole and the wavy background is). The header should be soldered from the SMD component side. Solder one pin first and check if the header sits flush with the PCB. If so, you could solder the other pins. If not, reheat the solder joint and adjust the header.
2. Solder the 2x8 female pin header next. Therefore, place the header on the SMD component side, flip the Main PCB over and solder from the wavy side. Again, solder one pin first and check if the header sits flush with the PCB. If so, you could solder the other pins. If not, reheat the solder joint and adjust the header. **Main PCB is done!**
3. Now it is time to start with the Control PCB. First place the 2x8 male pin header and the 2x3 male pin header on the SMD side of the Control PCB. Flip the PCB and solder the pins from the Component side.
4. Put one screw through the PCB from the component side, tighten the screw with the standoff from the SMD side.
5. Place the ten jack sockets, the three potentiometers, the push button, the three LEDs and the two sub-mini switches from the component side. The placement of each component is marked on the PCB. **Place but do not solder yet!** Mind the orientation of push button and the LEDs. Also take care that the ON-ON switch is placed at the cycle location and the ON-OFF-ON at the range location.
6. Screw on and tighten the knurled nuts from the jacks and the hex nuts from the potentiometers. Have a look if everything is aligned.
7. Now turn the module over and solder the Potentiometers and jack sockets from the SMD Side.
8. Next use something to keep the switches and push button in place and solder one pin first. Have a look if everything is still aligned and solder the rest of the pins.
9. Now it is time to solder the LEDs. Solder one after one and take care that they hit the translucent spot on the backside of the Frontpanel. Trim down the remaining legs of the LED.
10. Put the Main PCB and the Control PCB together, use another screw to fix the sandwich where the standoff is placed.

Congratulations, your build is done!

11. if it is needed to adjust the Shape, set the Range switch to its middle position, put the Module into cycle mode and push the push button to start the oscillation. Set Rise and Fall to fully CCW and the Shape knob into middle. Now you could adjust the shape via the small trimmer behind the square hole in the Main PCB. The shape should be displayed at a perfect triangle at the output.